

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

*I _N Akhilesh Kumar(192365074) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Nalluri Akhilesh Kumar
Signature of the student:

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty
- ii. Explain how A* would handle the same situation, including how it balances cost and heuristic
- iii. Analyze the impact of heuristic accuracy on both algorithms
- iv. Discuss how dynamic changes (blocked paths) affect search performance
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

Scenario

A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic because roads may suddenly become blocked, weather conditions change continuously, and movement costs vary depending on terrain. The drone has access to:

- A heuristic estimate (straight-line distance to the destination)
- Partial real-time updates about blocked paths
- Variable movement costs based on terrain and weather

The drone must choose the safest and fastest route while minimizing delay.

1. Behavior of Greedy Best-First Search (GBFS)

Greedy Best-First Search selects the next node based **only on the heuristic value ($h(n)$)**, which estimates how close the node is to the destination.

How it behaves in this scenario

- The drone always chooses the path that appears closest to the destination.
- It ignores the actual travel cost caused by floods, damaged roads, or bad weather.
- If a chosen route suddenly becomes blocked, the drone must search again from its current position.
- It quickly finds a route but does not guarantee that the route is the safest or shortest.

Strengths

- Very fast in emergency situations.
- Uses less memory than many other search algorithms.
- Suitable when a quick decision is more important than finding the optimal route.
- Easy to implement.

Weaknesses

- Does not consider the actual movement cost.
- Can choose risky or expensive paths.
- May get trapped in dead ends when obstacles appear.
- Does not guarantee the optimal solution.

Example

Suppose two possible paths exist:

- **Path A:** Straight line distance = 5 km, but flooded and slow.
- **Path B:** Straight line distance = 7 km, but clear and safe.

Greedy Best-First Search selects **Path A** because it has the smaller heuristic value, even though it may take much longer.

2. Behavior of A* Search

A* Search uses both the actual cost traveled and the estimated remaining distance.

The evaluation function is:

$$f(n) = g(n) + h(n)$$

Where:

- **$g(n)$** = Actual cost from the starting point
- **$h(n)$** = Estimated cost to the destination (heuristic)
- **$f(n)$** = Total estimated cost

How A* behaves in this scenario

- The drone considers both travel cost and estimated remaining distance.
- Flooded roads increase the movement cost.
- Dangerous weather also increases the path cost.
- When a blocked road is detected, A* recalculates another path with minimum total cost.

- It balances speed and safety.

Advantages

- Finds the optimal path if the heuristic is accurate.
- Avoids costly flooded or dangerous routes.
- Produces reliable navigation decisions.
- Suitable for dynamic environments with replanning.

Disadvantages

- Requires more computation than Greedy Search.
- Uses more memory because it stores many explored nodes.
- Replanning after frequent environmental changes increases computation time.

Example

Suppose:

Path	Actual Cost $g(n)$	Heuristic $h(n)$	Total $f(n)$
A	20	5	25
B	10	12	22

Although Path A appears closer, A* selects **Path B** because its total estimated cost is lower.

3. Impact of Heuristic Accuracy

The heuristic function plays a major role in the performance of both algorithms.

Accurate Heuristic

When the straight-line estimate closely matches the real distance:

Greedy Best-First Search

- Quickly reaches the destination.
- Makes fewer mistakes.
- Performance improves significantly.

A*

- Finds the optimal path efficiently.
- Expands fewer unnecessary nodes.
- Executes faster.

Poor Heuristic

When the heuristic estimate is inaccurate:

Greedy Best-First Search

- Frequently chooses incorrect paths.
- May repeatedly encounter blocked roads.
- Search performance decreases.

A*

- Still finds the optimal solution.
- Explores more nodes.
- Computation time increases.

Comparison

Heuristic Quality	Greedy Best-First Search	A* Search
Accurate	Very fast	Fast and optimal
Moderately accurate	Acceptable	Good
Poor	Often chooses wrong paths	Still optimal but slower

4. Effect of Dynamic Changes (Blocked Paths)

Floods create unpredictable road blockages.

Effect on Greedy Best-First Search

- May continue toward a blocked road because it only follows the heuristic.
- Frequently restarts the search.
- Wastes time exploring unsuitable paths.
- Performance decreases significantly in highly dynamic environments.

Effect on A*

- Updates the path cost when new information arrives.
- Recalculates the best alternative route.
- Handles changing environments more effectively.

- Maintains better route quality despite obstacles.

Comparison

Dynamic Situation	Greedy Best-First Search	A* Search
Blocked road	May restart completely	Replans efficiently
Weather changes	Usually ignored	Included in movement cost
Variable terrain	Ignored	Considered in cost calculation
Route quality	Unreliable	Reliable

5. Recommended Algorithm

Recommended: A* Search Algorithm

Justification

Although Greedy Best-First Search is faster, emergency medicine delivery requires **both speed and safety**. A* is more suitable because it considers the actual travel cost along with the estimated distance.

Reasons

1. Optimal Route

- Finds the minimum-cost path.
- Avoids unnecessary delays.

2. Considers Real Conditions

- Includes flooded roads.
- Includes weather conditions.
- Includes terrain difficulty.

3. Improved Safety

- Avoids risky or dangerous routes.
- Reduces the chance of drone failure.

4. Handles Dynamic Changes

- Can replan when roads become blocked.
- Adjusts to new environmental information.

5. Reliable Emergency Delivery

- Ensures medicines reach affected people quickly and safely.
- Balances efficiency with accuracy.

Final Comparison

Feature	Greedy Best-First Search	A* Search
Uses Heuristic	Yes	Yes
Uses Actual Cost	No	Yes
Speed	Very Fast	Moderate
Memory Usage	Low	Higher
Optimal Solution	No	Yes (with admissible heuristic)
Handles Dynamic Changes	Poor	Good
Suitable for Emergency Medicine Delivery	Partially	Highly Suitable

Conclusion

In a flood-affected region, the environment changes continuously because of blocked roads, varying terrain, and unpredictable weather. **Greedy Best-First Search** provides quick decisions but may choose unsafe or costly routes since it relies only on the heuristic estimate. In contrast, **A*** evaluates both the actual travel cost and the estimated remaining distance, enabling it to find safer and more efficient paths. Although A* requires more computation and memory, it delivers reliable and optimal routes, making it the **best choice for AI-powered emergency medicine delivery drones** where minimizing delay, ensuring safety, and adapting to real-time changes are critical.

2. **Scenario:** A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

- Explain how Hill Climbing would work in this scenario and identify its limitations
- Discuss issues like local maxima, plateaus, and ridges with real-world interpretation

- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations
- iv. Recommend the best approach and justify your choice based on scalability and adaptability

Scenario

A smart city uses an AI system to optimize traffic signal timings across hundreds of intersections. The objectives are to:

- Minimize total waiting time.
- Reduce fuel consumption.
- Decrease traffic congestion.

The system must continuously adjust traffic signals based on changing traffic flow. However, the search space is extremely large, traffic patterns change dynamically, and the system may become trapped in locally optimal solutions.

Formulating the Problem as an Optimization Problem

The traffic signal optimization problem can be modeled as an optimization problem where the AI searches for the best traffic signal timings that minimize congestion and travel delays.

Objective Function

The objective is to minimize:

- Total vehicle waiting time
- Fuel consumption
- Traffic congestion

This can be represented as:

Minimize:

$$\text{Cost} = \text{Waiting Time} + \text{Fuel Consumption} + \text{Congestion Level}$$

The AI system searches for the combination of signal timings that gives the lowest overall cost.

Decision Variables

The system adjusts:

- Green light duration
- Red light duration
- Yellow light duration
- Signal coordination between neighboring intersections

Constraints

- Minimum and maximum green signal duration.
- Emergency vehicles should receive priority.
- Pedestrian crossing time must be maintained.
- Traffic rules and safety regulations must be followed.

Why Traditional Search is Inefficient

Traditional search algorithms such as Breadth-First Search (BFS) and Depth-First Search (DFS) are not suitable because:

- The number of possible signal timing combinations is extremely large.
- Traffic conditions change every minute.
- Exploring every possible solution requires excessive computation time.
- Traditional search does not optimize continuously in real time.
- It cannot quickly adapt to unexpected traffic congestion or accidents.

Therefore, optimization algorithms are preferred over exhaustive search methods.

1. Hill Climbing Algorithm

Hill Climbing is a local search algorithm that starts with an initial traffic signal plan and continuously makes small improvements.

How Hill Climbing Works

1. Start with an initial traffic signal schedule.
2. Measure the total waiting time and congestion.
3. Slightly modify one or more signal timings.
4. If the new solution improves traffic flow, accept it.
5. Repeat the process until no further improvement is found.

Example

Suppose the current average waiting time is:

Current Waiting Time = 180 seconds

After increasing one green signal by 10 seconds:

New Waiting Time = 160 seconds

Since traffic improves, the AI keeps the new timing.

The algorithm continues improving until it cannot find a better neighboring solution.

Advantages

- Simple to implement.
- Fast execution.
- Requires less memory.
- Suitable for real-time optimization.

Limitations

- Can become stuck at local optimum solutions.
- Cannot always find the global best solution.
- Sensitive to the starting solution.
- Performs poorly when the search space is very complex.

2. Local Maxima, Plateaus, and Ridges

(a) Local Maxima (Local Optimum)

A local optimum is a solution that is better than its neighboring solutions but is not the best overall solution.

Real-World Interpretation

Suppose one intersection is optimized perfectly.

However,

Nearby intersections become congested.

Hill Climbing stops because small changes appear to worsen the current solution, even though a much better city-wide solution exists.

(b) Plateaus

A plateau is a flat area where many neighboring solutions have the same value.

Real-World Interpretation

Changing signal timing by a few seconds produces no noticeable improvement in traffic.

The algorithm cannot determine which direction is better.

It may stop or move randomly.

(c) Ridges

A ridge is a narrow path where improvement requires changing multiple variables simultaneously.

Real-World Interpretation

Improving traffic may require adjusting five nearby traffic signals together.

Changing only one signal does not improve traffic.

Hill Climbing changes only one variable at a time, so it cannot easily move along the ridge.

Summary Table

Problem	Meaning	Traffic Example
Local Optimum	Best nearby solution	One intersection improves while overall traffic remains poor
Plateau	No improvement in neighboring solutions	Changing timings gives nearly identical traffic flow
Ridge	Improvement requires multiple simultaneous changes	Several connected intersections must be adjusted together

3. Simulated Annealing and Genetic Algorithms

Simulated Annealing

Simulated Annealing is an improved version of Hill Climbing.

It sometimes accepts worse solutions temporarily to escape local optimum solutions.

How it Works

1. Start with an initial solution.
2. Generate a neighboring solution.
3. If it is better, accept it.
4. If it is worse, accept it with a certain probability.
5. Gradually reduce the probability of accepting worse solutions (cooling process).
6. Eventually settle on a high-quality solution.

Advantages

- Escapes local optimum solutions.
- Better exploration of the search space.
- Works well in dynamic optimization problems.
- Produces near-optimal solutions.

Genetic Algorithm (GA)

Genetic Algorithm is inspired by natural evolution.

Instead of improving only one solution, it works with an entire population of solutions.

Steps

1. Generate multiple traffic signal plans.
2. Evaluate each plan using a fitness function.
3. Select the best plans.
4. Perform crossover to combine good features.
5. Apply mutation to introduce diversity.
6. Repeat until a high-quality solution is obtained.

Advantages

- Explores many solutions simultaneously.
- Avoids getting trapped in local optimum solutions.
- Suitable for very large search spaces.
- Easily adapts to changing traffic conditions.
- Produces highly optimized signal timings.

Comparison

Feature	Hill Climbing	Simulated Annealing	Genetic Algorithm
Local Search	Yes	Yes	No (Population-based)
Escapes Local Optimum	No	Yes	Yes
Handles Large Search Space	Moderate	Good	Excellent
Supports Dynamic Environment	Moderate	Good	Excellent
Global Optimization	Poor	Good	Very Good

4. Recommended Approach

Recommended Algorithm: Genetic Algorithm

Justification

For a smart city with hundreds of intersections, the search space is extremely large and traffic conditions change continuously. Genetic Algorithms are the most suitable because they explore multiple solutions simultaneously and adapt well to dynamic environments.

Reasons

1. Highly Scalable

- Can optimize hundreds of intersections simultaneously.
- Performs well even in very large search spaces.

2. Avoids Local Optimum

- Population-based search prevents premature convergence.
- Mutation introduces new solutions.

3. Adaptive

- Responds effectively to changing traffic patterns.
- Can continuously improve signal timings.

4. Better Optimization

- Optimizes waiting time, fuel consumption, and congestion together.
- Produces high-quality solutions.

5. Parallel Processing

- Multiple candidate solutions can be evaluated simultaneously, making it suitable for modern smart city systems.

Final Comparison

Feature	Hill Climbing	Simulated Annealing	Genetic Algorithm
Search Strategy	Local Search	Probabilistic Local Search	Evolutionary Population Search
Escapes Local Optimum	No	Yes	Yes
Handles Plateaus	Poor	Good	Excellent
Handles Ridges	Poor	Better	Excellent
Scalability	Moderate	Good	Excellent
Adaptability	Moderate	Good	Excellent

Best for Smart Traffic System	No	Good	Yes
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Conclusion

Optimizing traffic signal timings in a smart city is a complex optimization problem because of the enormous search space and constantly changing traffic conditions. Traditional search algorithms are inefficient as they cannot explore all possible signal combinations or adapt quickly to real-time changes. **Hill Climbing** offers a simple and fast method for improving traffic flow but often becomes trapped in local optima, plateaus, or ridges. **Simulated Annealing** improves upon Hill Climbing by occasionally accepting worse solutions, helping it escape local optima. However, **Genetic Algorithms** provide the most effective solution by exploring multiple candidate signal plans simultaneously, avoiding premature convergence, and adapting efficiently to dynamic traffic conditions. Therefore, **Genetic Algorithms are the most suitable approach for large-scale smart city traffic optimization** due to their scalability, adaptability, and ability to produce high-quality solutions.

3. **Scenario:** An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i.Explain why this problem requires an online search agent instead of offline search
- ii.Analyze the challenges of partial observability and dynamic environments
- iii.Describe how the rover builds and updates its internal model
- iv.Compare online search with uninformed and informed search strategies
- v.Justify the most suitable approach considering uncertainty, adaptability, and safety

1. Why This Problem Requires an Online Search Agent Instead of Offline Search

An **offline search agent** assumes that the entire environment is known before the search begins. It computes a complete path from the start to the goal and then follows that path without changes.

In the Mars rover scenario, this approach is not practical because:

- The rover does not have a complete map of the Martian surface.
- New obstacles such as rocks or craters may be detected only when the rover gets closer.
- The terrain may be more difficult than expected, requiring the rover to change its route.

- Communication with Earth is delayed, so the rover must make decisions on its own.

An **online search agent** is more suitable because it explores the environment while searching. It observes the surroundings using onboard sensors, updates its knowledge, and selects the next action based on the latest information. This enables the rover to safely reach its destination even in an unknown environment.

Difference Between Offline and Online Search

Offline Search	Online Search
Requires a complete map before searching	Works without a complete map
Computes the entire path first	Searches while moving
Cannot easily adapt to new obstacles	Continuously adapts to new information
Suitable for known environments	Suitable for unknown environments like Mars

2. Challenges of Partial Observability and Dynamic Environments

(a) Partial Observability

Partial observability means the rover cannot see the entire environment at once because its sensors have a limited range.

Challenges

- Hidden rocks or craters may not be detected until the rover gets close.
- The rover cannot accurately predict what lies beyond its sensor range.
- Decisions must be made using incomplete information.
- There is a higher risk of selecting unsafe paths.

Example

A crater located behind a hill may not be visible initially. The rover detects it only after reaching the hill and must immediately choose an alternative route.

(b) Dynamic Environment

A dynamic environment is one that changes over time or presents new information while the agent is operating.

Challenges

- Loose sand may make travel slower than expected.
- Dust storms may reduce visibility.
- Newly detected obstacles may block the planned path.

- The rover must frequently revise its route.

Example

The rover plans to travel through an open area, but sensor readings later reveal large rocks. It must replan the route to avoid the obstacle.

3. How the Rover Builds and Updates Its Internal Model

The rover maintains an **internal model** of its surroundings using data collected from onboard sensors.

Step 1: Initial State

The rover starts with only its starting position and mission objectives.

Step 2: Environment Observation

Sensors such as cameras, lidar, radar, and proximity sensors collect information about:

- Rocks
- Craters
- Slopes
- Terrain type
- Safe paths

Step 3: Map Construction

The rover creates a local map by marking:

- Safe areas
- Dangerous areas
- Explored regions
- Unexplored regions

Step 4: Decision Making

The rover selects the next movement based on:

- Current location
- Newly discovered obstacles
- Estimated distance to the target
- Safety of available paths

Step 5: Continuous Updating

After every movement:

- New sensor data is collected.
- The map is updated.
- Previously unknown obstacles are added.
- The next decision is recalculated.

This cycle continues until all samples are collected.

4. Comparison of Online Search with Uninformed and Informed Search

(a) Uninformed Search

Uninformed search algorithms do not use heuristic information.

Examples:

- Breadth-First Search (BFS)
- Depth-First Search (DFS)
- Uniform Cost Search (UCS)

Characteristics

- Require knowledge of the search space.
- Explore many unnecessary states.
- Computationally expensive.
- Not suitable for unknown environments.

(b) Informed Search

Informed search algorithms use heuristic information to guide the search.

Examples:

- Greedy Best-First Search
- A* Search

Characteristics

- Faster than uninformed search.
- Uses estimated distance to the goal.
- Works well when a map is available.
- Performance decreases if the environment changes frequently.

(c) Online Search

Online search makes decisions while exploring.

Characteristics

- Does not require a complete map.
- Learns from experience.
- Continuously updates its knowledge.
- Adapts immediately to newly discovered hazards.
- Suitable for unknown and changing environments.

Comparison Table

Feature	Uninformed Search	Informed Search	Online Search
Requires Complete Map	Yes	Usually Yes	No
Uses Heuristic	No	Yes	May use heuristics with real-time data
Handles Unknown Environment	Poor	Limited	Excellent
Learns During Search	No	No	Yes
Adapts to New Obstacles	No	Limited	Yes
Suitable for Mars Rover	No	Partially	Yes

5. Recommended Approach

Recommended Approach: Online Search Agent

Justification

An Online Search Agent is the most suitable approach because the Mars rover operates in an unknown and unpredictable environment. It cannot rely on a precomputed path since new hazards are discovered only during exploration.

Reasons

1. Handles Uncertainty

- Works with incomplete information.
- Makes decisions even when the full environment is unknown.

2. Highly Adaptive

- Continuously updates the internal map.

- Replans routes whenever new obstacles are detected.

3. Ensures Safety

- Avoids dangerous terrain using real-time sensor information.
- Reduces the risk of accidents.

4. Real-Time Decision Making

- Makes immediate decisions without waiting for instructions from Earth.
- Suitable for missions with communication delays.

5. Efficient Exploration

- Explores only the necessary areas.
- Conserves battery power and computational resources.
- Increases the chances of successfully collecting scientific samples.

Final Comparison

Feature	Offline Search	Uninformed Search	Informed Search	Online Search
Complete Map Required	Yes	Yes	Usually Yes	No
Uses Heuristic	No	No	Yes	Optional
Handles Unknown Terrain	No	Poor	Moderate	Excellent
Real-Time Decision Making	No	No	Limited	Yes
Adaptability	Low	Low	Moderate	High
Safety	Moderate	Low	Good	Excellent
Best Choice for Mars Rover	No	No	Partially	Yes

Conclusion

The Mars rover exploration problem is a **real-time decision-making problem** because the rover operates in an unknown environment with limited sensing capability and continuously discovers new information. **Offline, uninformed, and informed search algorithms** are less suitable because they generally assume that the environment is known in advance and cannot easily adapt to unexpected hazards. An **Online Search Agent** is the most appropriate solution

because it explores while searching, continuously updates its internal map using sensor data, and replans its path whenever new obstacles are detected. This approach provides the highest level of **uncertainty handling, adaptability, safety, and autonomous decision-making**, making it ideal for Mars exploration missions.

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

- i. Clearly define variables, domains, and constraints with explanation
- ii. Explain how backtracking search works in this scenario
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency
- iv. Analyze real-world challenges and justify the best strategy for scalability

Formulating the Problem as a Constraint Satisfaction Problem (CSP)

A Constraint Satisfaction Problem (CSP) consists of three main components:

- **Variables**
- **Domains**
- **Constraints**

The goal is to assign a value from the domain to every variable such that all constraints are satisfied.

1. Variables, Domains, and Constraints

(a) Variables

Variables represent the exams that need to be scheduled.

Example:

- X_1 = Data Structures Exam
- X_2 = Artificial Intelligence Exam
- X_3 = Database Management Systems Exam
- X_4 = Operating Systems Exam
- X_5 = Computer Networks Exam

Each variable requires:

- A time slot
- An exam hall

(b) Domains

The domain represents all possible values that each variable can take.

For example:

Time Slots

- Morning (9:00–12:00)
- Afternoon (1:00–4:00)
- Evening (5:00–8:00)

Exam Halls

- Hall A
- Hall B
- Hall C
- Hall D

Example:

Domain(X_i)

Time:

{Morning, Afternoon, Evening}

Hall:

{A, B, C, D}

Every exam can initially be assigned to any available time slot and hall.

(c) Constraints

The timetable must satisfy several constraints.

Constraint 1: No Student Exam Overlap

A student enrolled in two subjects cannot have both exams at the same time.

Example:

If Student A is taking:

- AI
- DBMS

Then:

AI Time $\neq$ DBMS Time

Constraint 2: Limited Exam Halls

Only one exam can use an exam hall during the same time slot.

Example:

Hall A cannot simultaneously conduct:

- AI Exam
- Operating Systems Exam

Constraint 3: Subject Ordering

Some examinations must be conducted before others.

Example:

Programming Fundamentals must occur before Data Structures.

Therefore,

Programming Fundamentals Time < Data Structures Time

Constraint 4: Faculty Availability

Faculty members may only be available during certain periods.

Example:

Professor X is available only in the morning.

Therefore,

AI Exam must be scheduled during a morning session.

Constraint 5: Special Student Accommodation

Students requiring special accommodations must be assigned appropriate exam halls.

Example:

Hall D contains accessibility facilities.

Students with disabilities should have exams scheduled in Hall D whenever required.

Summary Table

Component	Example
Variables	Exams (AI, DBMS, OS, CN, etc.)
Domains	Time slots and exam halls
Constraints	No overlap, hall limits, faculty availability, subject order, special accommodations

2. Backtracking Search

Backtracking Search is a CSP-solving algorithm that assigns values to variables one by one. If a constraint is violated, the algorithm goes back (backtracks) and tries another assignment.

Steps

Step 1

Assign the first exam.

Example:

AI → Monday Morning → Hall A

Step 2

Assign the second exam.

DBMS → Monday Morning → Hall B

Check constraints.

If many students take both AI and DBMS,

Conflict occurs.

Step 3

Backtrack

Move DBMS to another time slot.

DBMS → Monday Afternoon → Hall B

Now all constraints are satisfied.

Continue assigning remaining exams.

Working Process

1. Select an unassigned exam.
2. Assign an available time slot and hall.
3. Check all constraints.
4. If constraints are satisfied, continue.
5. If a conflict occurs, undo the assignment.
6. Try another value.
7. Repeat until every exam is scheduled.

Advantages

- Guarantees a valid timetable if one exists.
- Simple and systematic.
- Works well for medium-sized scheduling problems.

Disadvantages

- May explore many unnecessary possibilities.
- Becomes slow when the number of exams is very large.
- Large universities increase computational complexity.

3. Constraint Propagation (Forward Checking)

Constraint propagation reduces the search space before conflicts occur.

One common technique is **Forward Checking**.

How Forward Checking Works

Whenever an exam is assigned, the algorithm removes conflicting values from the domains of remaining exams.

Example:

Suppose:

AI → Monday Morning

Then:

Every exam sharing students with AI cannot use Monday Morning.

That time slot is immediately removed from their domains.

This prevents future conflicts.

Example

Initial Domains

Subject	Available Time Slots
AI	Morning, Afternoon, Evening
DBMS	Morning, Afternoon, Evening
OS	Morning, Afternoon, Evening

Assign:

AI → Morning

Forward Checking updates the domains:

Subject Remaining Time Slots

DBMS Afternoon, Evening

OS Afternoon, Evening

The algorithm avoids assigning conflicting times later.

Advantages of Forward Checking

- Detects conflicts early.
- Reduces unnecessary backtracking.
- Shrinks the search space.
- Improves execution speed.
- Efficient for large CSP problems.

4. Real-World Challenges and Best Strategy for Scalability

Real-World Challenges

(a) Large Number of Courses

A university may have hundreds of subjects.

More exams increase the number of possible schedules.

(b) Large Student Population

Thousands of students may enroll in multiple subjects.

Avoiding timetable clashes becomes more difficult.

(c) Limited Resources

Only a limited number of:

- Exam halls
- Invigilators
- Faculty members

are available.

(d) Faculty Availability

Faculty may have different availability schedules.

The timetable must accommodate these restrictions.

(e) Special Accommodation Requirements

Some students require:

- Wheelchair-accessible halls
- Extra writing time
- Separate examination rooms

These increase the number of constraints.

(f) Last-Minute Changes

Unexpected events such as faculty illness, hall maintenance, or emergency situations may require the timetable to be modified quickly.

Best Strategy for Scalability

Recommended Strategy:

Backtracking Search with Constraint Propagation (Forward Checking)

Justification

This combination is highly effective for solving large CSPs because it combines systematic searching with early conflict detection.

Reasons

1. Efficient Search

Backtracking explores only feasible solutions.

2. Early Conflict Detection

Forward Checking removes invalid choices before they create conflicts.

3. Reduced Computation

The search space becomes much smaller.

4. Highly Scalable

Suitable for universities with hundreds of courses and thousands of students.

5. Flexible

Additional constraints, such as new faculty schedules or special accommodations, can be incorporated without redesigning the entire algorithm.

Final Comparison

Feature	Backtracking Only	Backtracking + Forward Checking
Conflict Detection	Late	Early
Backtracking Required	High	Low
Search Space	Large	Reduced
Execution Speed	Slower	Faster

Scalability	Moderate	Excellent
Suitable for University Timetabling	Good	Excellent

Conclusion

University exam timetable generation is a **Constraint Satisfaction Problem (CSP)** because exams must be assigned to time slots and halls while satisfying multiple constraints such as avoiding student exam clashes, respecting hall capacities, faculty availability, subject ordering, and special accommodation requirements. **Backtracking Search** systematically assigns exams and revises decisions whenever conflicts arise, ensuring a valid timetable if one exists. However, using **Constraint Propagation through Forward Checking** significantly improves efficiency by eliminating invalid choices early, reducing unnecessary backtracking and shrinking the search space. Therefore, **Backtracking Search combined with Forward Checking** is the most suitable strategy for large universities because it offers **high scalability, faster execution, efficient conflict handling, and the flexibility to accommodate real-world scheduling constrain**

5. Scenario: Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

- i.Explain how the Minimax algorithm models this problem
- ii.Analyze the computational challenges of large game trees
- iii.Explain how Alpha-Beta pruning improves efficiency with detailed reasoning
- iv.Design an evaluation function and justify the factors considered
- v.Recommend a strategy for balancing optimality and real-time performance

1. How the Minimax Algorithm Models This Problem

The **Minimax algorithm** is a decision-making algorithm used in two-player competitive games. It assumes that:

- The AI (MAX player) always tries to maximize its chances of winning.
- The Human opponent (MIN player) always tries to minimize the AI's chances of winning.

The algorithm builds a **game tree**, where:

- Each node represents a game state.
- Each branch represents a possible move.
- Leaf nodes represent the final or evaluated game positions.

The AI explores possible future moves and selects the move that provides the best guaranteed outcome, assuming the opponent also plays optimally.

Working of Minimax

1. Generate all possible moves from the current game state.
2. Simulate the opponent's possible responses.
3. Continue this process until a predefined search depth or terminal state is reached.
4. Evaluate the leaf nodes using an evaluation function.
5. Propagate the values upward:
 - **MAX nodes** choose the highest value.
 - **MIN nodes** choose the lowest value.
6. Select the move with the highest final value.

Example

Suppose the AI has three possible moves.

Move	Evaluation Score
Move A	8
Move B	5
Move C	10

The AI selects **Move C** because it gives the highest guaranteed score after considering the opponent's best responses.

2. Computational Challenges of Large Game Trees

In real-time strategy games, the number of possible moves increases rapidly.

(a) Huge Branching Factor

Each game state may have many possible actions.

Example:

- Build units
- Attack
- Defend
- Gather resources
- Upgrade buildings

If each state has 20 possible moves and the AI searches 6 levels deep:

Total nodes = $20^6 = 64,000,000$ nodes

Exploring every node is computationally expensive.

(b) Time Constraints

The AI often has only a few milliseconds to make a decision.

Searching the entire game tree may take too long, causing noticeable delays.

(c) Memory Requirements

Large game trees require significant memory to store states and evaluations.

This increases hardware requirements and reduces performance.

(d) Dynamic Game Environment

Human players may make unpredictable decisions.

The AI must frequently update its search and adapt to new situations.

(e) Real-Time Decision Making

The AI cannot spend several seconds computing the perfect move.

Instead, it must quickly choose a strong move within the available time.

3. How Alpha-Beta Pruning Improves Efficiency

Alpha-Beta Pruning is an optimization technique for the Minimax algorithm. It reduces the number of nodes that need to be evaluated without affecting the final decision.

Key Concepts

- **Alpha (α):** The best value that the MAX player can guarantee so far.
- **Beta (β):** The best value that the MIN player can guarantee so far.

If at any point $\alpha \geq \beta$, the remaining branches of that subtree cannot influence the final decision and are **pruned (ignored)**.

Working

1. Perform Minimax search.

2. Maintain alpha and beta values while exploring the tree.
3. If a branch cannot produce a better outcome than one already found, stop exploring that branch.
4. Continue searching only promising branches.

Example

Suppose the AI has already found a move with a value of **10**.

While exploring another branch, the opponent can force the value down to **6**.

Since **6** is worse than the already available value **10**, there is no need to explore the rest of that branch.

The remaining nodes are pruned, saving computation time.

Advantages

- Reduces the number of nodes evaluated.
- Produces the same optimal move as Minimax.
- Improves decision speed.
- Enables deeper searches within the same time limit.
- Makes real-time game AI more practical.

4. Evaluation Function

The evaluation function estimates how favorable a game state is for the AI.

A higher score indicates a stronger position for the AI, while a lower score favors the opponent.

Factors Considered

1. **Army Strength**
 - Number and power of units controlled by the AI.
2. **Resource Availability**
 - Amount of gold, food, energy, or other resources.
3. **Map Control**
 - Percentage of strategic locations or territory controlled.
4. **Health of Units**
 - Remaining health of important units and structures.

5. Defensive Strength

- Quality of defenses protecting the AI's base.

6. Enemy Strength

- Number and power of the opponent's units (reduces the score).

7. Objective Progress

- Progress toward achieving the game's objectives, such as capturing bases or completing missions.

Example Evaluation Function

Evaluation Score = (Army Strength + Resources + Map Control + Unit Health + Objective Progress) – Enemy Strength

A higher score represents a more advantageous game state for the AI.

5. Recommended Strategy for Balancing Optimality and Real-Time Performance

Recommended Strategy

Minimax with Alpha-Beta Pruning and Depth-Limited Search

Justification

This combination provides strong decision quality while keeping computation within real-time limits.

Reasons

1. Near-Optimal Decisions

Minimax evaluates future moves and chooses the best available strategy.

2. Faster Search

Alpha-Beta Pruning eliminates unnecessary branches, reducing computation.

3. Real-Time Performance

Depth-limited search restricts how far the AI explores, ensuring decisions are made within strict time limits.

4. Scalability

The algorithm remains effective even as the game complexity increases because pruning reduces the effective search space.

5. Better User Experience

Players experience intelligent AI behavior without noticeable delays or lag.

Final Comparison

Feature	Minimax	Minimax + Alpha-Beta Pruning
Finds Optimal Move	Yes	Yes
Evaluates All Nodes	Yes	No
Execution Speed	Slow	Fast
Memory Usage	High	Lower
Suitable for Real-Time Games	Limited	Excellent
Scalability	Moderate	High

Conclusion

Developing an AI opponent for a real-time strategy game requires making intelligent decisions under strict time constraints while exploring a vast number of possible moves. The **Minimax algorithm** provides a systematic way to model this problem by assuming that the AI maximizes its advantage while the human opponent minimizes it. However, the exponential growth of the game tree makes a full Minimax search computationally expensive. **Alpha-Beta Pruning** addresses this challenge by eliminating branches that cannot affect the final decision, significantly reducing the number of nodes that need to be evaluated without sacrificing optimality. By combining **Minimax, Alpha-Beta Pruning, a well-designed evaluation function, and depth-limited search**, the AI achieves an effective balance between **optimal decision-making, computational efficiency, scalability, and real-time performance**, making it the most suitable approach for modern real-time strategy games.